Viral Genomic Sequencing using Illumina Viral Surveillance Kit:

<u>Sample Extraction, Library Preparation, Loading an NGS Run and Run QC</u>

Table of Contents

1 PURPOSE

Viral whole genome sequencing is crucial for infectious diseases surveillance and prevention. This assay will provide genetic composition data of viruses of great public health risks, enabling identification of mutations that can influence critical factors of an outbreak, such as transmissibility, virulence and resistant to antiviral treatment or vaccines. The workflow uses universal total nucleic acid extraction method coupled with a hybrid-capture viral sequence enrichment to enable whole genome sequencing (WGS) of over 200 viral species via a single assay, while ensuring the completeness of genome coverage. When the sample provided is unextracted, the procedure includes a total nucleic acid extraction step using the Promega Maxwell RCS viral total nucleic acid purification kit. Next, viral nucleic acid is processed for WGS using Illumina DNA/RNA Prep with Enrichment and the Viral Surveillance Panel v2 library preparation kit.

The Promega Maxwell Viral Total Nucleic Acid Purification kit extracts both viral DNA and RNA content from diverse input sample types, including environmental samples and clinical samples stored in VTM, UTM and PBS. This is achieved by first mixing samples with lysis buffer to denature proteins and lipids, while releasing and stabilizing nucleic acid. The sample is loaded onto a cartridge containing paramagnetic beads that attract total nucleic acid. A magnetic plunger is then used to separate out bead-bound nucleic acid to allow wash and elution. This extraction method allows consistent and high-throughput automation that prepares 48 samples at a time.

Next, Illumina DNA/RNA Prep with Enrichment uses on-bead tagmentation followed by a single hybridization step to provide a rapid workflow for generating enriched libraries. The kit allows for different virus types, starting with either DNA extract, RNA extract that is converted to cDNA, or extract containing both RNA and DNA. The samples are tagmented and amplified to add indexes and adapter sequences. The resulting libraries are pooled for one to three-plex enrichment. During enrichment, the sequencing library is bound to biotinylated probes that target viral sequences and purified via streptavidin beads that are bound to magnetic stand. The extensive collection of probes target over 200 viruses identified as important risks to public health by the WHO. Probe-based enrichment eliminates the need for a high read depth that is required for shotgun metagenomics sequencing. Compared to other targeted sequencing methods, such as amplicon sequencing, the hybrid-capture method also provides more uniform coverage across genomes and a greater ability to identify mutations, making the VSP ideal for outbreak surveillance.

2 SCOPE

The AMD viral genomic sequencing protocol is intended for:

- Following One Health principles, to provide viral genomic surveillance data to epidemiologists and academic partners to monitor the spread and mutations of viral pathogens.
- To facilitate sufficient nucleic acid recovery directly from diagnostic specimens to enable next generation sequencing, across diverse viral conditions of public health importances.
- To enable the use of a single sequencing workflow for direct-from-specimen sequencing efforts on diverse viral species.
- To collaborate with national and global public health organizations to establish a standardized workflow for viral genomic sequencing via hybrid capture chemistry.

All personnel are required and expected to properly extract total nucleic acid, prepare a library, load a pooled library, correctly handle reagents, troubleshoot technical issues, and perform proper instrument maintenance to ensure optimal conditions on the Illumina sequencers.

3 REAGENTS / MEDIA / STANDARDS

3.1 Specimen Extraction Reagents

- Promega Maxwell Viral Total Nucleic Acid Purification Kit (48 reactions, CAT#AS1330; 144 reactions, CAT#ASB1330)
    - Lysis Buffer
    - Proteinase K (PK) Solution
    - Sterile Nuclease-free Water
- Molecular Grade Water as Mock

3.2 Sequencing Library Preparation Reagents

The Illumina Viral Surveillance Panel v2 kit, Set A (Illumina Catalog Number: 20108081) includes library prep and enrichment reagents (Illumina RNA Prep with Enrichment), an Enrichment Oligo Panel, and Index Adapters.

A. Illumina RNA Prep with Enrichment Kit (7 boxes)

1. Box 1 of 7: Illumina Purification Beads (IPBs), Store at 2°C to 8°C.

Quantity	Reagent	Description	Cap Color
3	IPB	Illumina Purification Beads	Clear (or Purple)

2. Box 2 of 7: Illumina DNA/RNA Prep – Tagmentation Buffers, Store at 2°C to 8°C.

Quantity	Reagent	Description	Cap Color
4	ST2	Stop Tagmentation Buffer 2	Red
1	TWB	Tagmentation Wash Buffer	Clear (or Purple)

3. Box 3 of 7: Illumina DNA/RNA Prep – Tagmentation Beads, Store at 2°C to 8°C.

Quantity	Reagent	Description	Cap Color
4	EBLTL	Enrichment Bead-Linked Transposomes	Yellow
2	RSB	Resuspension Buffer	Clear

4. Box 4 of 7: Illumina RNA Fast Hyb Enrichment Beads and Buffers, Store at 2°C to 8°C.

Quantity	Reagent	Description	Cap Color
1	EHB2	Enrich Hyb Buffer 2	Clear
1	ET2	Elute Target Buffer 2	Clear
1	RSB	Resuspension Buffer	Clear
4	SMB4	Streptavidin Magnetic Beads 4	Clear

5. Box 5 of 7: Illumina DNA/RNA Prep – Tagmentation PCR Reagents, Store at -25°C to -15°C.

Quantity	Reagent	Description	Cap Color
4	EPM	Enhanced PCR Mix	Clear
4	TB1	Tagmentation Buffer 1	Clear

6. Box 6 of 7: Illumina RNA Fast Hyb Enrichment PCR and Buffers, Store at -25°C to -15°C.

Quantity	Reagent	Description	Cap Color
2	EE1	Enrichment Elution Buffer 1	Clear

4	EEW	Enhanced Enrichment Wash	Amber
2	EPM	Enhanced PCR Mix	Clear
1	HP3	2N NaOH	Clear
1	NHB2	Hyb Buffer 2 + IDT NXT Blockers	Blue
1	PPC	PCR Primer Cocktail	Clear

7. Box 7 of 7: Illumina cDNA Synthesis, Store at -25°C to -15°C.

Quantity	Reagent	Description	Cap Color
4	EPH3	Elute, Prime, Fragment High Concentration Mix	Clear
4	FSA	First Strand Synthesis Act D Mix	Amber
2	RSB	Resuspension Buffer	Clear
1	RVT	Reverse Transcriptase	Clear
4	SMM	Second Strand Making Master Mix	Clear

B. Index Adapters - Illumina DNA/RNA UD Indexes (96 Indexes, 96 Samples), Store at -25°C to -15°C.

Quantity	Reagent	Description
1	UDP0001 – UDP0096	Set A Index Adapter Plate

C. Enrichment Oligo Panel - Viral Surveillance Panel v2, Store at -25°C to -15°C. **Can be ordered separately.**

Panel Name	Description	Illumina Catalog Number
VSP v2 Panel	Viral Surveillance Panel v2	20107489

D. Ethanol, molecular grade, 95 – 100% (Fisher Catalog Number: BP2818-500 or equivalent)
E. Isopropanol, lab grade, 70% or equivalent for disinfection purposes (Fisher Catalog Number: 04-335-309 or equivalent)
F. Water, molecular grade (Fisher Catalog Number: BP28191 or equivalent).
G. Qubit Reagents (Thermo Fisher Catalog: Q32851)
    1. dsDNA HS Reagent (Component A). Store at Room Temperature.
    2. dsDNA HS Buffer (Component B). Store at Room Temperature.
    3. dsDNA HS Standard #1: Store at 2°C - 8°C.
    4. dsDNA HS Standard #2: Store at 2°C - 8°C.
H. High Sensitivity DNA Screen Tape Analysis Kit:
    1. High Sensitivity D5000 ScreenTape (Agilent Technologies Catalog Number: 5067-5592).
    2. High Sensitivity D5000 Reagents (Agilent Technologies Catalog Number: 5067-5593).
I. MiSeq Reagent Kit: Box 1 of 2. Store at -25°C – 15°C.
    1. Hybridization Buffer (HYB-1/HT-1)
    2. MiSeq Cartridge
        - V2 300 Cycles – Illumina, Catalog Number: MS-102-2002
J. MiSeq Reagent Kit: Box 2 of 2. Store at 2°C – 8°C.
    1. Incorporation Buffer (PR2)

   2. Flow Cell

K. Sodium Hydroxide, 1N (Sigma-Aldrich, Catalog Number: S2770-100mL or equivalent). pH should be ≥ 12.5
   1. Preparations in advance of testing: Make 100µL aliquots of 1N NaOH and freeze. Store for a length of time not to exceed the expiration of the stock NaOH. **Freshly diluted (0.2N) NaOH should be used within 6 hours of dilution and kept on ice.**

L. Sodium Hypochlorite, 4 – 4.99% (Sigma-Aldrich, Catalog Number: 239305-25mL or equivalent)

M. Tween 20 (Sigma-Aldrich, Catalog Number: P9416-50mL)

N. 10nM Phi-X Library (Illumina Catalog Number: FC-110-3001). Store at -25°C – 15°C.

O. 1M Tris-HCL, pH 7.0 (Fisher Scientific Catalog Number BP1756-100 or equivalent).

4 SUPPLIES / MATERIALS

A. Greiner Bio-One LUMITRAC Binding 96-Well Polystyrene Microplates (Greiner Catalog Number: 07-000-129)

B. Solution basins, sterile (Fisher Scientific Catalog Number: 13-681-504 or equivalent)

C. 96 well PCR plates, skirted, hard shell low profile, thin-wall (BioRad Catalog Number: HSP-9601 or equivalent)

D. Eppendorf twin.tec PCR Plate 96, semi-skirted, 250µL, PCR clean (Eppendorf Catalog Number: 951020303 or equivalent)

E. Microseal B adhesive seal (BioRad Catalog Number: MSB-1001 or equivalent)
   Equivalent option: Microseal F adhesive foil seal (BioRad Catalog Number: MSF-1001 or equivalent)

F. Deep well storage “MIDI” plates, 96 well (Fisher Catalog Number: AB-0859 or equivalent)

G. Ice or cold blocks

H. Illumina MiSeq wash tray (Two per instrument; recommended one for bleach washes, one for water washes only)

I. Lint-free wipes (Fisher Scientific Catalog Number: 06-666 or equivalent)

J. Lens paper (Fisher Scientific Catalog Number: 11-996 or equivalent)

K. MiSeq wash tubes (Illumina Catalog Number: MS-102-9999)

5 EQUIPMENT

A. Promega Maxwell CSC 48 Instrument

B. Molecular Devices SpectraMax M2 Microplate Reader

C. Thermal cycler with Deep Well module, capable of accepting a 96-well plate, with heated lid.

D. Microplate centrifuge

E. Vortex

F. Agilent TapeStation 4200 System (G2991BA)

G. Magnetic Stand-96 (ThermoFisher Catalog Number: AM10027 or equivalent)

H. Micropipettes, capable of volumes from 1µL to 1000µL. Single and multichannel (20µL, 200µL, 300µL, and 1000 µL volumes).

Note: **Two sets of pipettes are suggested; one for working with pre-amplified products and reagents, and one set for working with post-PCR amplified products and reagents.**

I. Plate shaker set to 1600rpm for 1 minute.

J. Microcentrifuge for quick spins.

K. Ice buckets/containers.

L. Illumina MiSeq™ System
   1. MiSeq™ Control Software v. 4.1.0.656
   2. Generate FASTQ v. 3.1.0.35
   3. Local Run Manager v. 3.0.0.3127
   4. MiSeq unique identifiers that communicate which runs were performed on them:
      - MiSeq M8782: “Taylor”

M. Heat block

N. Hybex Microsample Incubator with the HeatBlock 32 well for 1.7mL tubes (ThermoFisher Catalog Numbers: NC9555274 and NC9710520)

O. ***Optional:*** UPS back-up for the MiSeq (Recommended: Staples, Cyberpower AVR Series Line Interactive 1.5kVA UPS, Catalog Number: IM1M140118)

P. ***Optional:*** External encrypted hard drive or server for data transfer and storage if BaseSpace or networking of the instrument is not available (Suggested: CDW, DataLocker H350 Basic Hard Drive 1TB USB 3.0, Catalog Number: 4075102 – This lab currently uses the "Virtual Machine" on server).

6 SPECIAL SAFETY PRECAUTIONS

A. Follow all safety procedures as outlined in the current version of the WAPHL Biosafety Manual (MediaLab Document #: 69790.17) and Safety and Health Manual (MediaLab Document #: 69790.38). These include:
   1. Appropriate gloves
   2. Lab coat
   3. Safety glasses
   4. Spill response
   5. Dispose of all chemical components in accordance with local and federal safety standards.

B. Dispose of all chemical components in accordance with local and federal safety standards (**AMD Risk Assessment Media Lab Document #: 69790.2678**):
   1. Ethanol: GHS Flammability Category 2. Keep away from heat sources.
   2. ST2: GHS Category 2A for eye damage/irritant.
   3. TB1: GHS Category 4 for acute toxicity (dust/mist), Category 2A for eye irritant and Category 1B for reproductive toxicity and carcinogenicity. Contains N,N=Dimethylformamide.
   4. EPM: GHS Category 4 for acute oral toxicity and Category 1 for specific organ toxicity. Contains tetramethylammonium chloride.
   5. HP3: GHS Category 1 Sub-category A for skin corrosion and irritation and Category 1 for serious eye damage and eye irritation.
   6. NHB2: GHS Category 2 for carcinogenicity and specific target organ toxicity when repeated exposure. Category 1B for reproductive toxicity.
   7. SMB4: GHS Category 2 for carcinogenicity and specific target organ toxicity when repeated exposure. Category 1B for reproductive toxicity.
   8. Qubit Reagents:
      - dsDNA HS Reagent (Component A): GHS Category 4 for flammable liquids.
      - dsDNA HS Buffer (Component B): Not a hazardous substance or mixture.
      - dsDNA HS Standard #1: Not a hazardous substance or mixture. Store at 2°C - 8°C.
      - dsDNA HS Standard #2: Not a hazardous substance or mixture. Store at 2°C - 8°C.
   9. Formamide: GHS Category 2 for carcinogenicity and for specific target organ toxicity; and Category 1B for reproductive toxicity.

Note: **For mitigation measures for the previously mentioned chemicals, please refer to AMD Risk Assessment document in MediaLab: 69790.2678.**

7 SPECIMEN/SAMPLE INFORMATION

A. Approved specimen sources
   1. For molecular specimen handling to prevent specimen loss, alteration, or contamination, please review the following documents in MediaLab:
      - Laboratory Contamination Prevention and Monitoring: Nucleic Acid Testing (MediaLab Document #: 69790.1229)
      - General Etiquette for Molecular Room Usage (MediaLab Document #: 69790.3085)
   2. Submitters should provide either nucleic acid extraction or clinical/environmental specimen, including but not limited to nasal/buccal/nasopharyngeal swab in VTM/UTM/PBS, urine or stool samples.
   3. Sample proceeding to library preparation must be separately evaluated to contain detectable viral genome via digital droplet PCR, quantitative PCR or BioFire detection. All viral genome detection/quantification performed prior to library preparation and sequencing will be performed using validated procedures.

4. If nucleic acid extraction is provided, proceed directly to library preparation step without Promega Maxwell nucleic acid purification.
    - A minimum volume of 20 uL of total nucleic acid is required. The sample must be shipped on dry ice and stored at -80°C or below.
    - Minimum volume of 2 mL is required for samples in VTM/UTM/PBS.
    - Minimum volume of 20 mL is required for urine.
    - Minimum volume of 10 mL is required for stool.

B. Storage

1. Prior to submission to AMD, DNA extracts should be stored at -20°C or below.
2. Prior to submission to AMD, total nucleic acid and RNA extracts should be stored at -80°C or below.
3. Upon extracting/receiving DNA, RNA or total nucleic acid, the samples should be stored at -80°C or below in AMD until sequencing.
4. Specimens to be extracted can be shipped on cold packs and stored at 2-8C if processing occurs within 72 hours of collection. Specimens should be shipped on dry ice and stored at -80C if processing occurs > 72 hours of collection.

C. Sample retention policy

1. Original specimens submitted by internal labs should be maintained by the originating laboratory following corresponding sample retention policies.
2. Leftover unextracted specimen should be stored in -80°C or below for at least 1 year.
3. Extracted nucleic acid should be aliquoted into > 10uL aliquots and stored in -80°C or below for at least 6 months.

D. Turn Around Time (TAT)

1. Surveillance sequence: TAT will be established between submitting lab, AMD lab, and EPI team. Surveillance sequencing will be tested and analyzed following WAPHL protocol. Epidemiologists will indicate the urgency of high-priority specimens.

E. Rejection Criteria

1. Epidemiologists will be notified of any specimen rejection or retest.
2. The samples received for sequencing will be analyzed by ddPCR/qPCR/BioFire prior to submission. If the samples are below the limit of detection by ddPCR/qPCR/BioFire for intended virus, the sample is not sequenced.
3. If the sample is previously analyzed using BioFire, a positive result is required.
4. If the sample is previously analyzed using quantitative RT-PCR/PCR, a virus-specific Ct value ≤ 25 is preferred, a Ct value ≤ 30 is acceptable.
5. If the sample is previously analyzed using ThermoFisher TaqMan open array assay, a virus-specific CRT value ≤ 15 is preferred, a CRT value ≤ 20 is acceptable.
6. If the sample is previously analyzed using droplet digital PCR, a viral genome concentration ≥ 10^4 genomic copies/µL preferred, ≥ 10^3 genomic copies/µL is acceptable.

F. Sample Quality

1. If input samples are previously extracted, the nucleic acid must be stored at -80°C with minimum number of freeze-thaw cycles.
2. If input samples are not previously extracted, the samples must be stored 2-8 °C for no more than 72 hours or kept at ≤ -80°C for long-term storage.

8 QUALITY

8.1 Proficiency Testing

- **Type**: ☐External Provider ☒Alternative PT, document #: **69790.4408 – Whole Genome Sequence Alternative Proficiency Testing Guidelines**

- **Provider**: Internal assessment consisting of 6 specimens is also part of the Proficiency Testing for the AMD lab.

- **Frequency**: Twice a year.

8.2 Quality Control

A. **New Reagent (Lot or Shipment) QC:**

1. Sample Extraction Reagents:

When the reagents are received by an AMD staff, a "Needs QC" label will be put on all the boxes from the kit, as well as the date received. Once the sequencing run is completed and analyzed, and the positive control sample passes the analytical QC criteria, the reagent kit will be considered QC'd. The run will be assessed by the AMD Lead and Supervisor for review.

2. Library Preparation Reagents:

- Illumina Viral Surveillance Panel v2 kit consists of 7 reagent boxes, 1 index box, and 1 enrichment oligo viral surveillance panel. As such, all arrive as a set. Together, one set allows up to 96 samples or 32 hybridization reactions to be prepped. Each individual box of a given 9-box set is to be labeled with a unique internal alphanumeric indicator (e.g. "Set A", "Set 3", etc.) and a colored sticker. This allows easy identification of individual boxes corresponding to a set, and to distinguish one set from another. Each library prep should be performed using a single library prep set to the extent possible. Mixing library prep kits, **<u>even if individual reagents share the same lot numbers</u>**, is discouraged. A reagent set may only be combined with a reagent set of the **<u>same lot number with Lead approval</u>**. **<u>Reagents of different lots are not to be combined</u>**, per AMD guidelines. When selecting a library prep set, personnel should select the set with the earliest expiration date.
- For library preparation QC, the reagent box from the kit (RNA Prep + UD Indexes + Enrichment Oligo Panel) with the shortest expiration date will be taken as the main lot number and receive date. When the kits are received by AMD staff, a "Needs QC" label will be put on all the boxes from the kit and the indexes that will go with it, as well as the date received. Once the sequencing run is finished, results will be analyzed in the respective analysis pipeline and if the positive control results come back matching the expected results, the lot will be considered QC'd. The run will be assessed by the AMD Lead and Supervisor prior to reporting to the submitting laboratories or epidemiology partners.

3. Sequencing Reagents:

When the kits are received by AMD staff, a "Needs QC" label will be put on each of the boxes, as well as the date received. Once the initial sequencing run using a new reagent lot is finished, results will be analyzed in the respective pipeline, and if the positive control results come back matching the expected result, and the sequencing metrics come back within range, the lot will be considered QC'd. The run will be assessed by the AMD Lead or Supervisor prior to reporting to the submitting laboratories or epidemiology partners. MiSeq reagents boxes should be selected to ensure that the boxes correspond to each other (i.e. were received on the same date).

C. Dual Purpose Control (DPC):

Each library prep includes a Dual-Purpose Control (DPC). This control is composed of synthetic viral genome with 1000 genomic copies per µL. This control will act as a positive control and will be measured with the rest of the samples through library fragment size estimation and quantification, and through bioinformatic analysis. The DPC is also used to confirm reagents are performing as expected.

DPC ID	Organism / Control Name	Reagent Source	Catalog Number	NCBI Reference Sequence Accession Number
1	Influenza seasonal (H1N1)	Twist	103001	NC_026438, NC_026435, NC_026437, NC_026433, NC_026436, NC_026434, NC_026431, NC_026432
2	Influenza seasonal (H3N2)	Twist	103002	NC_007373, NC_007372, NC_007371, NC_007366, NC_007369, NC_007368, NC_007367, NC_007370
3	Influenza seasonal (IBV)	Twist	103003	NC_002204, NC_002205, NC_002206, NC_002207, NC_002208, NC_002209, NC_002210, NC_002211
4	CoV-229E	Twist	103011	NC_002645.1

5	CoV-NL63	Twist	103012	NC_005831.2
6	CoV-OC43	Twist	103013	NC_006213.1
7	SARS-CoV-2 Control 2	Twist	102024	MN908947.3
8	SARS-CoV-2 Control 48	Twist	105204	EPI_ISL_6841980
9	SARS-CoV-2 Control 66	Twist	106198	EPI_ISL_12454576
10	Mumps virus	Twist	103007	NC_002200.1
11	Measles virus	Twist	103009	NC_001498.1
12	Human parainfluenza virus 1	Twist	103008	NC_003461.1
13	Human parainfluenza virus 4	Twist	103010	NC_021928.1

D. QC Parameters for Library Prep: Accurate quantification and proper quality check of NGS libraries is important to achieve a successful sequencing run.

1. **Library quantification:**
Libraries are quantified through the Qubit™ dsDNA HS Assay Kit. The kit enables quick and selective detection of low and high abundance DNA in samples and specifically detects double stranded DNA. The instruments used to quantify the individual libraries and the pooled libraries, are the SpectraMax and the Qubit™ 2.0 Fluorometer. Consult with a Lead Microbiologist as needed.

2. **Fragment Size (also known as insert size):**
The second library quality check is fragment size. The fragment size is the average length of the DNA to be sequenced. Fragment size impacts the quality of the sequencing run and data output. Illumina RNA Prep is designed to generate a peak size of approximately 250 – 350bp (insert size of 100 – 200 bp), but larger fragment sizes are sometimes observed. In practice, fragment sizes of 250 – 650 bp are optimal for sequencing. If library fragment size cannot be ascertained, it is acceptable to use a predicted fragment size of 300 bp. <u>Inability to measure library fragment size is not an issue that requires work stoppage</u>. The instrument used to measure the insert size is the TapeStation 4200. The purpose of the TapeStation is to define the average fragment size which is used to calculate the molarity of the pooled library. ***Note:*** Shorter fragments cluster more efficiently than longer fragments during sequencing. Right- or left-skewed libraries can affect the quality of the sequencing run (i.e. causing diminished Q30 and/or focus errors).

3. **Negative Control:**
To confirm that consumables used during the run are not contaminated, and to guarantee that no large-scale cross-contamination between samples takes place, a No Template Control (NTC, or negative control) will be included in each run. The NTC will consist of water, going through the same process as a sample would. If the NTC comes back at the end of library preparation with a concentration below $1^{ng}/_{\mu L}$, the library would be considered clean, and the library can proceed with Part II of the sequencing protocol. Clean NTC will be sequenced. On the other hand, if the concentration obtained is greater than $1^{ng}/_{\mu L}$, the library would be considered contaminated and should not proceed with sequencing.

Library Preparation – NGS Wet Lab QC Parameters				
	Pass		Fail	
Library Prep QC	Outcome	Action	Outcome	Action
***Concentration > 4nM** ***Library size > 250bp** ***NTC < 1ng/µL**	DNA Library detected, pass QC quantification metrics	Proceed through normalization and loading the instrument	DNA library does not pass QC (i.e. no library detected, low concentration)	Repeat library preparation

E. PhiX Spike-in control: Phi-X is added to every sequencing run at the desired concentration with an expected margin of error of 1 – 1.5%.

1. PhiX control v3 is a well-characterized, adapter-ligated library used as a control for sequencing runs.
2. Provides quality control for cluster generation, sequencing, and alignment.
3. Serves as a calibration control for crosstalk matrix generation, pre-phasing, and phasing.
4. In the event of an issue during the WGS run, PhiX can be used to evaluate run and instrument performance to aid in troubleshooting.
5. For MiSeq runs, PhiX is added at a concentration of 1%.

F. **Reagents:**
   1. Store all reagents and RNA/DNA on ice while working on the bench unless indicated otherwise.
   2. Lot numbers of all the reagents and run metrics are entered into a reagent tracking log. Performance of lots can be monitored by comparing run metrics over time.
   3. Reagents with the earliest expiration dates should be prioritized for use, to avoid waste.

G. **Library Prep Workbook:**
   Library Prep workbook is designed to perform all calculations necessary to successfully create a library. However, always be aware that even though the workbook is calculating correctly, it is recommended to double check that all calculations make sense and/or have not been lost while building the run.

9 CALIBRATION

Preventive maintenance and calibrations follow the manufacturer's recommendations. These are performed by the vendor or a qualified service provider. Service reports are reviewed and filed by the supervisor for records retention. A post preventative maintenance verification is performed on all the Illumina instruments.

10 PROCEDURE

Part I: Specimen Processing and Nucleic Acid Extraction

10.1 Specimen Processing

- Receive specimens from Virology or ELS upon confirmation on the presence of viral genomes using validated assays.
- Store specimens at appropriate temperature:
  - Store unextracted specimens at 2-8°C if extraction is planned within 72 hours post collection. Otherwise, store specimens at -80°C or colder.
  - If specimen is extracted using other validated methods, store extracts at -80°C until library preparation, omit the Promega Maxwell extraction steps (**Section 10.2**).
- Label sterile 1.7 mL Eppendorf tubes with corresponding specimen numbers and set aside for extraction use.
- Pipette 200 µL of each specimen into labeled sterile 1.7 mL Eppendorf tubes.
- Store the leftover unextracted specimens in a 2.0 mL cryovial tube at -80°C or colder.
- Add 200 µL of Lysis Buffer + 20 µL Proteinase K to the 1.7 mL Eppendorf tube containing specimen.
- Touch vortex for 10 – 15 seconds, then incubate at room temperature for 10 minutes.

10.2 Maxwell CSC 48 Extraction

Note: **Change gloves before handling cartridges , RSC plungers, and Elution Tubes to avoid contamination.**

A. **Maxwell Cartridge Preparation**
   1. Place the cartridges to be used in the deck tray(s) with well # 1 (the largest well in the cartridge) facing away from the elution tubes.
   2. Press down on the cartridge to snap it into position and carefully peel back the seal so that all plastic comes off the top of the cartridge. (Ensure that all sealing tape and any residual adhesive are removed before placing the cartridges in the instrument).
   3. Place one plunger into well #8 of each cartridge.
   4. Place an empty elution tube into the elution tube position for each cartridge in the deck tray(s).
   5. Add 50uL Nuclease-Free Water to the bottom of each elution tube.
   6. Transfer total volume (~420uL) of sample lysate to well #1 (the largest well of the cartridge).

   Additional Note:

   - Specimen or reagent spills or any part of the deck tray should be cleaned with an Opticide-soaked rag and allowed to sit for 3 minutes before wiping.
   - Use only the 0.5 mL Elution Tubes provided in the kit; other tubes may be incompatible with the Maxwell Instrument.

B. Performing Maxwell CSC 48 Extraction

1. Select "Start" Option on tablet screen to begin.
2. Scan or enter Maxwell kit bar code.
3. Select cartridge positions and enter sample tracking information for both trays. Press **Proceed** to start. You will be prompted that the door will open.
4. Add cartridges and press down firmly to snap the cartridges in place.
5. Remove seals from cartridges and insert elution tubes.
6. Place a plunger into cartridge well #8 for each cartridge position.
7. Transfer samples, reagents and elution buffer for each extraction as directed by the extraction kit technical manual.
8. Place the deck trays into the instrument.
9. Verify proper deck tray setup and select start on the tablet PC screen to initiate the vision system scan.
10. The system will identify any errors in deck tray setup.
11. Press the red exclamation mark to identify errors and correct the problem.
12. Once the method runs are completed, select Open Door on the tablet PC screen.
13. Cap elution tubes, remove deck trays and discard cartridges according to site procedures.
14. Results will be displayed for completed runs in the Report view. See Operating manual TM623 for instructions and cleaning procedures.
15. Aliquot the elution into 8-strip tubes containing 3 aliquots of 12.5 µL of each extraction and store at -80C or lower.

Part II: Sequencing Library Preparation using Illumina DNA/RNA Prep with Viral Surveillance Panel V2 Enrichment kit

10.3 Preparation for Protocol

Before starting this procedure, pre-program the thermal cycler with the following programs:

A. Denature RNA (**VSP_Den_RNA**): Total run time is ~5 minutes.
   1. Preheat the lid at 100°C.
   2. Reaction volume: 17µL.
   3. 65°C for 5 minutes.
   4. Hold at 4°C.

B. Synthesize First Strand cDNA (**VSP_FSS**): Total run time is ~43 minutes.
   1. Preheat lid set at 100°C.
   2. Reaction volume: 25µL.
   3. 25°C for 10 mnutes.
   4. 42°C for 15 minutes.
   5. 70°C for 15 minutes.
   6. Hold at 4°C.

C. Synthesize Second Strand cDNA (**VSP_SSS**): Total run time is ~1 hour.
   1. Preheat lid at 40°C.
   2. Reaction volume: 50µL.
   3. 16°C for 1 hour.
   4. Hold at 4°C.

D. Tagment DNA (**VSP_TAG**): Total run time is ~5 minutes.
   1. Preheat lid at 100°C.
   2. Reaction volume: 50µL.
   3. 55°C for 5 minutes.
   4. Hold at 10°C.

E. Amplification of Tagmented DNA (**VSP_TAG_PCR**): Total run time is ~50 – 60 minutes.
   1. Preheat lid at 100°C.
   2. Reaction volume: 50µL.
   3. 72°C for 3 minutes.

4. 98°C for 3 minutes.
5. 17 cycles of:
    i. 98°C for 20 seconds.
    ii. 60°C for 30 seconds.
    iii. 72°C for 1 minute.
6. 72°C for 3 minutes.
7. Hold at 10°C.

F. Hybridize Probes (**VSP_HYB**): Total run time is ~115 minutes.
1. Preheat lid at 100°C.
2. Reaction volume: 25µL.
3. 98°C for 5 minutes.
4. 18 cycles of 1 minute each:
    i. 94°C for the first cycle.
    ii. Decrease 2°C per subsequent cycle.
5. 58°C for 90 minutes.
6. Hold at 58°C for ≤ 24 hours.

G. Capture Hybridized Probes **(VSP_INC**): Use the incubator setting. Make sure to use the deep well module for this incubation.
1. Preheat lid at 70°C.
2. Reaction volume: 100µL.
3. Hold at 58°C.

H. Amplify Enriched Library (**VSP_AMP**): Total run time is ~35 minutes.
1. Prehaet lid at 100°C.
2. Reaction volume: 50µL.
3. 98°C for 30 seconds.
4. 14 cycles of:
    i. 98°C for 10 seconds.
    ii. 60°C for 30 seconds.
    iii. 72°C for 30 seconds
5. 72°C for 5 minutes.
6. Hold at 10°C.

10.4 Library preparation

A. Prepare for Protocol
1. Avoid extended pauses until RNA is converted into double-stranded cDNA.
2. Avoid freeze-thawing samples that will be used for sequencing.
3. Keep RNA on ice before and during the preparation until cDNA is formed.
4. Prepare reagents as follows:

Reagent	Box Name	Instructions
IPB	Illumina Purification Beads	Bring to room temperature.
EPH3	Illumina cDNA Synthesis	Thaw at room temperature.
FSA	Illumina cDNA Synthesis	Thaw on ice.
RSB	Illumina cDNA Synthesis	Thaw at room temperature.
RVT	Illumina cDNA Synthesis	Thaw on ice.
SMM	Illumina cDNA Synthesis	Thaw on ice.

B. Denature RNA
1. Keep RNA samples on ice until ready to start.
2. Inside the AirClean, add 8.5 µL of RNA sample or total nucleic acid into each sample well of a PCR plate.
3. Add 8.5 µL of EPH3 to each sample well.
4. Gently mix by pipetting up and down for a minimum to 10 times.
5. Seal the plate with Microseal 'B' and centrifuge for 3 seconds at 280g.

6. Place the 96-well plate in the thermal cycler and run the program VSP_Den_RNA. The program takes about ~5 minutes to finish.
7. While the program is running, ensure FSA and RVT reagents are thaw and kept cold. Invert to mix, **DO NOT VORTEX, and quickly spin down**.
8. Once the thermal cycler is in the 4°C hold, centrifuge the plate for 10 seconds at 280g.

C. Synthesize First Strand cDNA

1. Prepare the master mix for the synthesis of the first cDNA strand per sample. Checklist formula already calculates overage.
   - FSA (9µL)
   - RVT (1µL)
2. **Do not vortex the master mix**. Pipette up and down to mix a minimum of 10 times.
3. Remove the seal from the PCR Plate and add 8µL of master mix to each well.
4. Pipette up and down for a minimum of 10 times to mix.
5. Centrifuge the plate for 10 seconds at 280g.
6. Place the 96-well plate in the thermal cycler and run the program VSP_FSS. The program takes about ~43 minutes to finish.
7. While waiting for the VSP_FSS program to finish:
   - Ensure the RSB and IPB are thawed. Vortex, invert to mix, and quickly spin down.
   - Ensure the SMM is thawed and has been kept on ice. Invert to mix and quickly spin down. **DO NOT VORTEX.**
   - Prepare 80% EtOH from fresh 100% ethanol.

D. Synthesize Second Strand cDNA

1. Once the VSP_FSS program has finished, centrifuge the 96-well plate for 10 seconds at 280g.
2. Invert the SMM to mix, and then quickly spin down. **DO NOT VORTEX**.
3. Add 25µL of SMM to each well. Mix by pipetting up and down for a minimum of 10 times. Seal the plate with Microseal 'B'.
4. Centrifuge the 96-well plate for 10 seconds at 280g.
5. Place the 96-well plate in the thermal cycler and run the VSP_SSS program. The program should run for about an hour.
6. Once the program is done, take the plate off the thermal cycler and centrifuge for 10 second at 280g.
7. Take the Illumina Purification Beads (IPBs) and resuspend well. It can take up to 1 minute. Vortex and invert mix as necessary.
8. Add 90µL of IPB to each well.
9. Seal the plate with Microseal 'B' and shake for 1 minute at 2200rpm.
10. Incubate at room temperature for 5 minutes. Centrifuge the plate for 10 seconds at 280g.
11. Place on the magnetic stand and wait until the liquid is clear, ~ 5 minutes.
12. Remove and discard the supernatant.
13. Wash the beads as follows:
    - Keep the plate on the magnetic stand and add 175µL fresh 80% EtOH to each well.
    - Wait 30 seconds.
    - Without disturbing the beads, remove and discard all supernatant.
14. Repeat wash a second time.
15. Using a 10µL pipette, remove and discard all residual EtOH.
16. Air-dry on the magnetic stand for 2 minutes. **Do not over-dry the beads.**
17. Remove from the magnetic stand.
18. Add 19.5µL RSB to each well. Seal the plate with Microseal 'B' and shake for 1 minute at 2600rpm. **Hold the plate in place.**
19. Incubate at room temperature for 2 minutes. Centrifuge the plate for 10 seconds at 280g.
20. Place the 96-well plate on the magnetic stand and wait until the liquid is clear, ~ 4 minutes.
21. Transfer 17.5µL of the supernatant to a new PCR plate. **Small amounts of bead carryover do not affect performance.**
22. Return all reagents back in their respective boxes at the correct temperature.

SAFE STOPPING POINT: Seal the plate with Microseal 'B' and store at -25°C to -15°C for up to 7 days.

E. Tagment DNA

Note: **If the DNA was stored in the freezer, take the plate out to thaw on ice or a cold block. Proceed to quickly vortex the plate and centrifuge to spin down for 10 seconds at 280g.**

1. Prepare reagents as follows:

Reagent	Box Name	Instructions
ST2	Illumina DNA/RNA Prep – Tagmentation Buffers	Bring to room temperature.
TWB	Illumina DNA/RNA Prep – Tagmentation Buffers	Bring to room temperature.
EBLTL	Illumina DNA/RNA Prep – Tagmentation Beads	Bring to room temperature.
EPM	Illumina DNA/RNA Prep – Tagmentation PCR Reagents	Thaw on ice or cold block.
TB1	Illumina DNA/RNA Prep – Tagmentation PCR Reagents	Bring to room temperature.
Index Adapter Plate	Illumina DNA/RNA UD Indexes Set A	Bring to room temperature.

2. In a 1.7mL tube, prepare the Tagmentation Master Mix per sample. Checklist already calculates overage.
   - TB1 (11.5µL)
   - EBLTL (11.5µL)
   - Molecular Grade Water (14.5µL)
3. Thoroughly vortex the Tagmentation Master Mix to resuspend.
4. Add 32.5µL of the master mix to each well.
5. Mix thoroughly by pipetting up and down for a minimum of 10 times. Seal the plate with Microseal 'B'.
6. Place in the thermocycler and run the program VSP_TAG, ~ 5 minutes.
7. During the incubation, ensure ST2 is fully warmed to room temperature and no precipitation is formed. Aliquot ST2 into strip tubes.
8. Once the plate reaches 10°C, remove the plate from the thermal cycler and centrifuge for 10 seconds at 280g.
9. Incubate the plate at room temperature for 2 minutes.
10. Add 10µL of ST2 to each well. Seal the plate with Microseal 'B' and shake for 1 minute at 2200rpm.
11. Incubate plate at room temperature for 5 minutes. Centrifuge the plate for 10 seconds at 280g.
12. Place on the magnetic stand and wait until the liquid is clear (~3 minutes).
13. Remove and discard all supernatants. Proceed to wash the beads:

Note: **Sample cross contamination is prone to happen at the TWB wash steps. Avoid foaming during pipetting.**

   - Remove from the magnetic stand.
   - Add 100µL of TWB to each well.
   - Seal the plate with Microseal 'B' and shake for 1 minute at 2000rpm. Alternatively, manually pipette to resuspend the beads.
   - Centrifuge the plate for 3 seconds at 280g.
   - Place on magnetic stand and wait until the liquid is clear (~3 minutes).
   - Remove and discard all supernatant.
14. Repeat the process a second time.
15. Wash beads a third time, but do not remove and discard the supernatant. Keep the plate on the magnetic stand, with the beads covered by the TWB to prevent them from over-drying.
16. Prepare the PCR Master Mix. The workbook calculations already include overage.
   - EPM (23µL) – **DO NOT vortex this reagent**.
   - Molecular Grade Water (23µL)
17. Invert to mix the Master Mix a minimum of ten times and quickly spin down. Aliquot into an appropriate reservoir.
18. Remove and discard the supernatant from the plate. Go back in with a p20 pipette and remove any remnants of TWB.
19. Remove the plate from the magnetic stand and immediately add 40µL of the PCR Master Mix to each well.
20. Using new pipette tips, pierce the foil covering the wells in the index adapter plate.
21. Add 10µL of the UD indexes into each well.
22. Set the pipette to 30µL, pipette up and down to mix for a minimum of ten times. Seal the plate with Microseal 'B' and place on the thermal cycler.

23. Run the VSP_TAG_PCR program. The program should take between 50 – 60 minutes.

SAFE STOPPING POINT: Seal the plate with Microseal 'B' and store at -25°C to -15°C for up to 7 days.

F. Clean Up Library

***Note:* If the DNA was stored in the freezer, take the plate out to thaw on ice or a cold block. Proceed to quickly vortex the plate and centrifuge to spin down for 10 seconds at 280g.**

1. Prepare reagents as follows:

Reagent	Box Name	Instructions
IPB	Illumina Purification Beads	Bring to room temperature.
RSB	Illumina DNA/RNA Prep – Tagmentation Beads	Bring to room temperature.

2. Prepare fresh 80% EtOH from absolute EtOH.
3. Take the plate out of the thermal cycler and centrifuge for 10 seconds at 280g.
4. Place on the magnetic stand and wait until the liquid is clear (~3 minutes).
5. Transfer 45µL of the supernatant from each well to a MIDI plate.
6. Vortex IPBs thoroughly and add 81µL to each well containing a sample.
7. Seal the plate with Microseal 'B' and shake for 1 minute at 2200rpm.
8. Incubate the plate for 5 minutes at room temperature.
9. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
10. Remove and discard all supernatant.
11. Wash beads as follows:
    - Keep on magnetic stand and add 175µL of fresh 80% EtOH to each well.
    - Wait 30 seconds.
    - Remove and discard the supernatant.
12. Wash beads a second time.
13. With a p20 pipette, remove all residual EtOH.
14. Air-dry the beads on the magnetic stand for 2 minutes. **Do not over dry the beads.**
15. Remove the plate from the magnetic stand and add 17µL of RSB to each well. Seal the plate with Microseal 'B' and shake for 1 minute at 2700rpm.
16. Incubate at room temperature for 2 minutes.
17. Place on the magnetic stand and wait until the liquid is clear (~ 2 minutes).
18. Transfer 15µL of the supernatant to a new PCR plate.

SAFE STOPPING POINT: Seal the plate with Microseal 'B' and store at -25°C to -15°C for up to 30 days.

G. Pool Library

***Note:* If the DNA was stored in the freezer, take the plate out to thaw on ice or a cold block. Proceed to quickly vortex the plate and centrifuge to spin down for 10 seconds at 280g.**

1. Quantify the individual libraries using the Qubit dsDNA HS Assay to ensure successful library generation.
   Note: Prior to hybridization, a successful sequencing library generated from clinical or environmental samples should have > 10 ng/µL concentration, whereas the sequencing library for NTC and DPC could have < 1 ng/µL concentration, due to little to no genomic diversity.

2. Pool undiluted libraries into one well of a PCR plate based on the sample volumes in the following table. If sample has low viral load, consider 1-plex reaction to maximize the chance of capturing the viral genome. Consult with an AMD Lead or Supervisor when designing the hybridization pooling strategy.

Library Pool Plexity	Each Pre-Enriched Library Volume (µl)	Total Volume (µl)
1-plex	7.5	7.5
3-plex	2.5	7.5

Note: The Illumina protocol pools pre-enriched libraries for one-plex, or three-plex enrichment. To achieve optimal performance, only pool pre-enriched library samples prepared by the same user, do not mix lots, and same index adapter plate.

H. Hybridize Probes

1. Preheat the microheating system to 50°C.
2. Set the thermal cycler to 50°C incubation, with heated lid off.
3. Prepare the following consumables:

Reagent	Box Name	Instructions
EHB2	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
NHB2	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
Viral Surveillance Panel v2	Viral Surveillance Panel v2	Thaw on ice

- EHB2—Vortex to mix. If crystals and cloudiness are observed, repeat vortex, or pipette up and down to mix well until the solution is clear.
- NHB2—Vortex to mix. Lay in the preheated microheating system and incubate for 5 minutes. Then vortex 3 times for 10 seconds each time. Pipette NHB2 to resuspend fully, make sure no precipitate is observed in the pipette tip. Then keep at 50°C until use.
- Illumina Viral Surveillance Panel v2—Vortex to mix then thaw on ice.

4. Rest the PCR plate **on the thermal cycler** set at 50°C, add the following reagents to each well of a new PCR plate *in the order listed*.

Note: Act fast! Do not let the reaction fall to room temperature as precipitates can form and hybridization will fail.

- One/Three-plex pre-enriched library pool (7.5 µl)
- NHB2 (12.5 µl)
- Illumina Viral Surveillance Panel v2 (2.5 µl)
- EHB2 (2.5 µl)

1. Pipette 10 times to mix, and then seal the plate with Microseal 'B'.
2. Centrifuge for 3 seconds at 280g.
3. Place on the preprogrammed thermal cycler and run the VSP_HYB program.
4. Allow to incubate at 58°C for 90 minutes to 24 hours.

Recommended Stopping Point: Allow the reaction to incubate overnight at 58°C for 18 – 24 hours to ensure maximum probe capture.

I. Capture Hybridized Probes

Note: Make sure to use the thermal cycler with deepwell module to ensure efficient heating. Act fast! Do not let the reaction fall to room temperature as precipitates can form and hybridization will fail.
Also, due to the small elution volume, use the Permagen magnet for future steps.

1. Prepare the following consumables:

Reagent	Box Name	Instructions
ET2	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
SMB3	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
EE1	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
EEW	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
EPM	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Thaw on ice.
HP3	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Bring to room temperature.
PPC	Illumina RNA Fast Hyb Enrichment Beads and Buffers	Thaw on ice.

2. Preheat the microheating system to 58°C.
3. Set the thermal cycler with deepwell modeule to incubator function at 58°C, preheat lid option and set to 70°C.
4. Prepare the following consumables:
   - EE1—Pipette to mix.
   - EEW—Vortex to mix. Place the tube on its side on the microheating system at 58°C.

- ET2—Vortex to mix.
- HP3—Vortex to mix.
- SMB4—Vortex and invert to mix.

5. The tube remains on the heat block until the wash procedures.

Capture

6. Centrifuge the sample plate at 280 × g for 10 seconds.
7. Vortex SMB4 to fully resuspend, ensure it has reached room temperature.
8. Add 62.5 µl SMB4 to each well.
9. Slowly pipette until the beads are resuspended and then seal.
10. Place the sample plate in the 58°C thermal cycler, close the lid, and incubate for 15 minutes. Keep the thermal cycler running continuously through the capture and four washes.
11. Centrifuge the sample plate at 280 × g for 10 seconds.
12. Place on a magnetic stand and wait until the liquid is clear (~2 minutes).
13. Remove and discard all supernatant from each well.

Wash

14. Remove from the magnetic stand.
15. Add 50 µl preheated EEW (amber tube) to each well. Seal and shake the plate at 2600 rpm for 2 minutes.
16. Return unused EEW to the microheating system and keep heated.
17. Place the sample plate on the thermal cycler and incubate for 5 minutes at 58°C.
18. Centrifuge at 280 × g for 3 seconds.
19. Place the plate on a magnetic stand and wait until the liquid is clear (~2 minutes).
20. Remove and discard all supernatant from each well.
21. Repeat steps 13–19 for a total of **three** EEW washes.

Transfer Wash

22. Remove the plate from the magnetic stand.
23. Add 50 µl preheated EEW (amber tube) to each well. Seal and shake the plate at 2600 rpm for 1 minute.
24. Seal and centrifuge at 280 × g for 3 seconds.
25. Transfer 50 µl to a new set of wells on a PCR plate.
26. Seal and centrifuge at 280 × g for 3 seconds.
27. Place the sample plate on the thermal cycler and incubate for 5 minutes at 58°C.
28. Place on a magnetic stand and wait until the liquid is clear (~2 minutes).
29. Remove and discard all supernatant from each well.
30. Use a 20 µl pipette to remove and discard residual liquid from each well.

Elution

31. Combine the following volumes to prepare an Elution Master Mix. Multiply each volume by the number of samples being processed.
    - EE1 (28.5 µl)
    - HP3 (1.5 µl)
32. Additional reagent is included in the volume to ensure accurate pipetting due to the potential of reagent foaming.
33. Pipette the Elution Master Mix to mix and then set aside at room temperature.
34. Remove the sample plate from the magnetic stand.
35. Add 23 µl Elution Master Mix to each well of the PCR plate. Seal plate and shake at 2600 rpm for 1 minute.
36. Incubate the plate at room temperature for 2 minutes.
37. Centrifuge at 280 × g for 10 seconds.
38. Place on a magnetic stand and wait until the liquid is clear (~2 minutes).
39. Transfer 21 µl supernatant from the PCR plate to the corresponding well of a new 96-well PCR plate.
40. Add 4 µl ET2 to each well containing 21 µl supernatant.
41. Set pipette to 20 µl and slowly pipette each well 10 times to mix.

42. Seal the plate and centrifuge the sample plate at 280 × g for 30 seconds.

J. Amplify Enriched Library

1. Invert PPC to mix, then briefly centrifuge. Add 5 µl PPC to each well of the PCR plate.
3. Invert EPM to mix, then briefly centrifuge. Add 20 µl EPM to each well.
4. Seal and shake at 2000 rpm for 1 minute.
5. Centrifuge at 280 × g for 10 seconds.
6. Place on the preprogrammed thermal cycler and run the VSP_AMP program.

K. Clean Up Enriched Library

1. Prepare reagents as follows:

Reagent	Box Name	Instructions
IPB	Illumina Purification Beads	Bring to room temperature.
RSB	Illumina DNA/RNA Prep – Tagmentation Beads	Bring to room temperature.

2. Prepare fresh 80% EtOH from absolute EtOH.
3. Take the plate out of the thermal cycler and centrifuge for 10 seconds at 280g.
4. Vortex & invert IPB several times ensuring resuspension.
5. Add 90 µL IPB to each well.
6. Seal the plate with Microseal 'B' and shake for 1 minute at 2200rpm.
7. Incubate the plate for 5 minutes at room temperature.
8. Centrifuge the plate for 10 seconds at 280g.
9. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
10. Remove and discard all supernatant.
11. Wash beads as follows:
    - Keep on magnetic stand and add 175µL of fresh 80% EtOH to each well.
    - Wait 30 seconds.
    - Remove and discard the supernatant.
12. Wash beads a second time.
13. With a p20 pipette, remove all residual EtOH.
14. Air-dry the beads on the magnetic stand for 2 – 3 minutes. **Do not over dry the beads.**
15. Remove the plate from the magnetic stand and add 32µL of RSB to each well.
16. Seal the plate with Microseal 'B' and shake for 1 minute at 2600rpm.
17. Incubate at room temperature for 2 minutes.
18. Centrifuge the plate for 10 seconds at 280g.
19. Place on the magnetic stand and wait until the liquid is clear (~ 2 minutes).
20. Transfer 30µL of the supernatant to a new PCR plate - This is the final product.

L. Quality Control and Pooling for Enriched Library

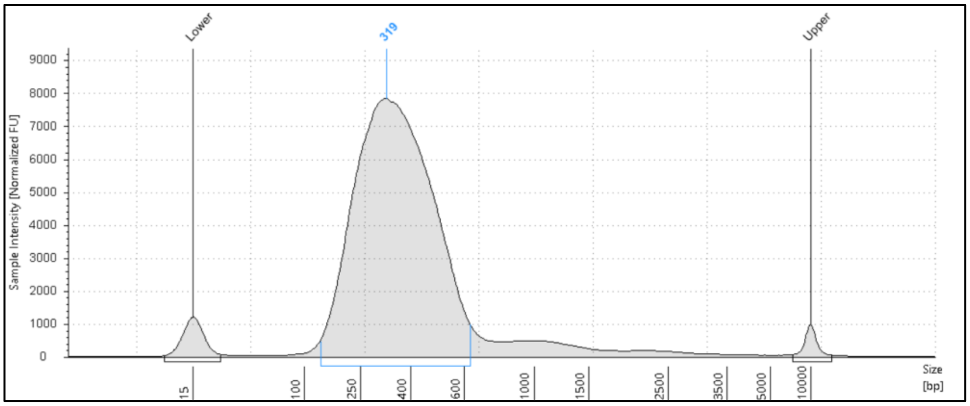
Library quantification and fragment size analysis metrics are specified in the quality control section.

1. Quantify the individual hybridization pools using Qubit dsDNA HS Assay Kit and the SpectraMax microplate reader, based on **MediaLab SOP 69790.956 WAPHL DNA Quantification and QC**.
2. Calculate the pooling ratio of individual library pools to make a final loading pool.
    - Based on the individual pool concentrations, adjust the normalized volume to add, so that the total ng amount of each pool is equivalent.
    - Combine individual pools together based on the normalized volume to make the final loading pool. For Example:

Pool#	Pool Concentration (ng/uL)	Normalized Volume to Add (uL)	Amount added (ng)
1	6.81	7.34	50
2	7.13	7.02	50
3	9.17	5.45	50

3. Determine the concentration and the fragment size of the final loading pool.
    - Quantify the final loading pool using Qubit dsDNA HS Assay Kit and the SpectraMax microplate reader.

- Determine the fragment size distribution of the finally loading pool using the High Sensitivity D5000 DNA screen tap and reagents on Tapestation 4200.
- The final library fragment size is the size of the main peak. For example, the following trace snapshot is an example of the typical VSP library. This library shown below has a fragment size of **319 bp**:



Part IV: Loading an NGS Run

10.5 Loading an NGS Run on MiSeq

A. **Step 1**: **Preparing the MiSeq for a Run**

1. Ensure there is enough space on the D:\ drive to support a run (at least 100GB). Delete older run files from both folders, MiSeq Output and MiSeq Analysis folders to free up space:
    i. Go to D:\Illumina\MiSeqOutput\ and delete any folders excluding the most recent folders.
    ii. Go to D:\Illumina\MiSeq Analysis\ and delete any folders excluding the most recent folders.
2. The sample sheet should be uploaded to the MiSeq at this time. Refer to MediaLab Document #: 69790.4364 for information on how to create and upload a sample sheet.

B. **Step 2: Preparing the Sequencing Reagents**

1. Plan enough time for thawing the cartridge and note that the HT-1 (HYB) buffer is in the same box as the cartridge. They must be thawed on ice or at 2°C – 8°C. See the table below for guidance regarding reagent kit storage after thawing:

Method of Thawing	Time Required to Thaw	Storage Life (2°C – 8°C)
Room Temperature Water Bath (Package opened)	~ 1 hour	24 hours
2°C – 8°C	8 – 12 hours	3 days

2. Remove the flow cell from 2°C – 8°C and allow it to equilibrate to room temperature for at least 10 minutes.

Note: **The sequencing cartridge can never be re-frozen.**

3. Once thawed, invert the reagent cartridge 10 times to mix, and then visually inspect to verify that all positions are completely thawed and free of precipitates.
If precipitates do not re-suspend, Call Illumina Technical Support (1-800-809-4566) for guidance.
4. Gently tap the cartridge on the bench to reduce air bubbles in the bottom of the reagent wells and ensure that the foil seals covering the wells are unobstructed.
5. Place the thawed reagent cartridge and HT-1 (HYB) buffer aside at 2°C - 8°C until the library is ready to be loaded.

C. **Step 3: Denaturation of the Pooled Amplified Libraries (PAL)**

1. Add 400 µL of molecular water to the 100 µL aliquot of 1N NaOH to reach the 0.2N NaOH working concentration. Vortex to mix.
2. Label a new empty tube as DAL (Denatured Amplified Library).

3. To the DAL labeled tube, add 5µL of the pooled library, and then 5µL of 0.2N NaOH. Vortex to mix and quickly spin down.
4. Incubate at room temperature for 5 minutes.
5. Once the 5-minute incubation is done, add 990µL of chilled HT-1 (HYB) buffer, and place the tube on ice.

5uL of 6nM pooled libraries + 5uL of NaOH. Incubate for 5 minutes. Add 990uL of HT1 = 30pM library pool

D. **Step 4: <u>Diluting to the final loading concentration</u>**

Depending on multiple factors (insert size, instrument, user, and chemistry) the denatured pooled library needs to be diluted before loading into cartridge. Illumina recommended loading concentrations for each sequencing chemistry are as follows:

Sequencing Chemistry	Recommended Loading Concentration for VSP Library	The loading concentration may be increased or decreased to help achieve desired cluster density.
V2	**10pM – 15pM**	**In general, over-clustering can lead to a decrease in quality while under-clustering can lead to low coverage and repeats. For VSPv2 library, 12 pM is a good starting point.**

Loading concentration is determined empirically and based on experience. Use the Reagent Tracker to determine a reasonable concentration or ask the AMD lead for guidance.

1. In a new tube labeled “DAL 2” (Loading Denatured Amplified Library), dilute the 30pM library pool to the desired loading concentration following the formula on the following box from the “Pool” tab:

Dilute 30 pM library pool to final loading concentration	
Desired Final Loading concentration (10 pM to 30 pM):	**15**
Volume of 30pM library pool (uL):	500
Volume of HT1 buffer (uL):	500

Note: **The initial PAL dilution concentration has been increased from 4nM to 6nM to allow for optimization of cluster density.**

E. **Step 5: <u>Phi-X Spike-In</u>**

1. Prepare the 30pM Phi-X stock following MediaLab Document #: 69790.3108.
2. Dilute the 30pM Phi-X stock to the concentration of the pooled library (for example, if a 15pM library will be loaded, dilute the Phi-X to 15pM).

Dilute PhiX to final loading concentration	
Desired final PhiX concentration (10 pM to 30 pM):	**15**
PhiX spiked concentration (%):	**1**
Available diluted PhiX volume + extra (uL):	11
Volume of 30pM PhiX (uL):	5.5
Volume of HT1 buffer (uL):	5.5

3. Remove 10 µL of denatured library from the “DAL 2” tube and discard.
4. Add 10 µL of the diluted Phi-X to the “DAL 2” tube. Invert mix and spin down.
5. Place on ice if stopping or go directly to the final steps.

F. **Step 6: <u>Cleaning the flow cell</u>**

1. Place paper towels or Wypalls in the bottom of a bin.
2. Place KimWipes, lens paper and 70% ethanol or an alcohol wipe within reach.

3. Remove the flow cell from the bottle and inspect it for any damage. Look for cracks or scratches in the glass. Inspect the ports and gasket (the gasket should be flush with plastic casing).
4. Using molecular grade water in a squirt bottle, gently spray down the flow cell, rinsing off all areas (glass and plastic portions very gently).
5. Dry off only the plastic areas gently with a KimWipe.
6. Use lens paper to dry the glass portion of the flow cell. If smudges persist or drying is difficult, repeat with water. If the problem persists, use an alcohol wipe or 70% ethanol on a piece of lens paper to gently clean the glass.

***Note:* Be very careful not to get ethanol on the two ports at the base of the flow cell. Set on lens paper until ready to load into the instrument**.

G. **Step 7: Heating the Loading DAL**

1. Place the "DAL 2" tube in the 96°C (±1°C) heat block for 2 minutes.
2. Immediately after, place the "DAL 2" on ice for at least 5 minutes. Do not go beyond 30 minutes.

H. **Step 8: Loading and starting a run**

1. Touch the "Sequence" button on the MiSeq Control Software main screen.
2. Select "Local Run Manager". Authenticate using the lab account, **micromolecularlab**. Press Next.
3. Check the box for Basespace Hub, and select the bottom option, "Run analysis, collaboration, and storage". Press Next.
4. Log in to Illumina using the micromoleculararb@doh.wa.gov account. Password will be kept in the lab space. Press Sign in, then Next.
5. Under the "Available Workgroup" screen, select "Personal". Press Next.
6. In the "Run Selection" screen, select your run under the "Available Runs" menu and verify the parameters match. Double check sample information by clicking "Preview Samples". If everything looks good, press Next.
7. Load the clean flow cell. Take this opportunity to look over the stage inside the flow cell chamber to see it is clean and dry.
8. Press down the latch. Check the screen to see that the RFID tag has been successfully recognized and press "Next".
9. Invert the Incorporation Buffer a couple of times, remove the lid and load it in place of the wash bottle. Lower the sipper and check the screen to see that the RFID tag has been successfully recognized and press "Next". Make sure the waste container in the MiSeq is empty.
10. Gently tap the cartridge on the bench to remove air bubbles in the reagent reservoirs.
11. Use a clean p1000 pipette tip to gently pierce the foil over the reservoir on the cartridge labeled "Load Samples" (Position 17).
12. At an angle and as close to the bottom of the reservoir as possible, gently load 600µL of the "DAL 2" into position 17.
13. Gently but firmly tap the cartridge on the counter to dislodge any air bubbles in the sample reservoir.
14. Remove the wash cartridge and replace it with the sample cartridge.
15. Check the screen to see that the RFID tag has been successfully recognized and press 'Next".
16. A run summary will appear on the screen. Look it over carefully and make any changes or correct any errors before starting the run.
17. Select "Next" on the load cartridge screen.
18. The pre-run check will begin. The flow check is the last to pass. Lab space can be cleaned, and equipment/reagents put away, but remember to come back to the MiSeq occasionally to check on the flow rate pre-run check.
19. Once the flow rate check passes, press "Start Run".

***Note:* Occasionally the flow rate check will fail. When this happens, 99% of the time it is due to the flow cell not seated properly in the chamber. Open the chamber and inspect the flow cell. Confirm that the gasket is flush and undamaged or distorted. Reseat the flow cell and select "Retry". Most of the time this settles the matter, and the flow rate check passes. If the flow rate check fails again, refer to MediaLab Document #: 69790.3109.**

I. **Step 9: <u>Post run instrument and data maintenance</u>**

1. Upon completion of a sequencing run, select “Start Wash”.
2. On the next screen, select “Perform optional template line wash”.
3. Perform a bleach wash followed by a maintenance wash as described in MediaLab Document #: 69790.3106.
4. Dispose of accumulated formamide waste inside the chemical fume hood (R-22) into the satellite container.
5. While the instrument is washing, transfer the fastq files stored on the MiSeq to the shared drive (Common2) on the Virtual Machine for long term storage.
6. Data is found in the D drive → **Illumina** → **MiSeq Output** → **Run Folder** (look for your run file) → **Alignment_1** → **Run Folder** (i.e. date of fastq generation/run end date) → **Fastq**. Within the Fastq folder the raw read data is in fast1.gz. format. Do not move the Undetermined, and Summary files.
7. Open a new file explorer window and navigate to the **Common2 drive** (<u>\\DOHIFLSEA04.dohinst.lcl</u> – usually mapped as the Z:\ drive).
8. Navigate the following pathway and create a folder using the run number.
    i. Common2 drive → **Shared Data** → **Fastq files** → **Year** (20xx) → **Run Folder**
9. Copy the fastq.gz files from the D:/ drive on the MiSeq to the run folder in the Common2 drive.

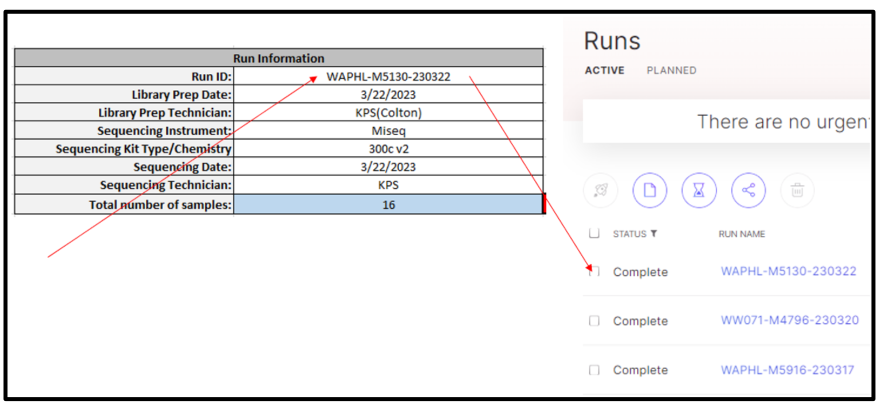
Note: **Transferring fastq.gz files is crucial after a run. These files are routinely deleted from the MiSeq as part of the routine maintenance.**

Part III: Data QC and Reporting

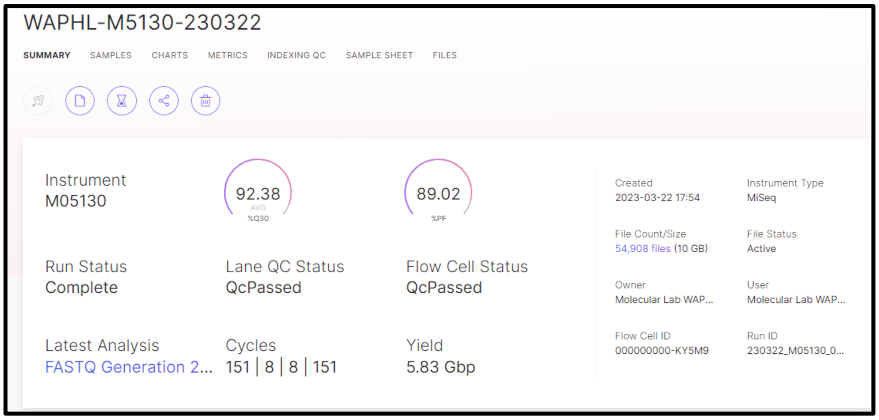
A report should be generated to capture relevant QC metrics for every run, whether it passes quality checks or not. Use BaseSpace to fill out the Metrics tab of the WGS run worksheet per the instructions below and deliver to the lead microbiologist as part of the run packet.

A. Log into BaseSpace, navigate to the Runs tab.

B. Look for your run ID under Run Name.

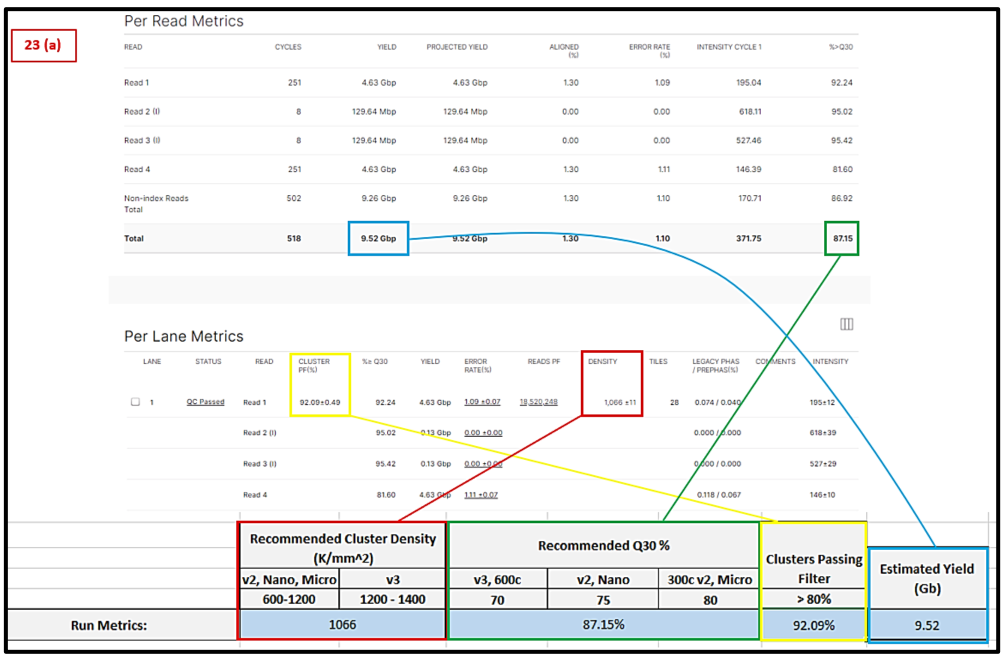


C. Click on the Run Name, and you will be redirected to the Summary of the run:

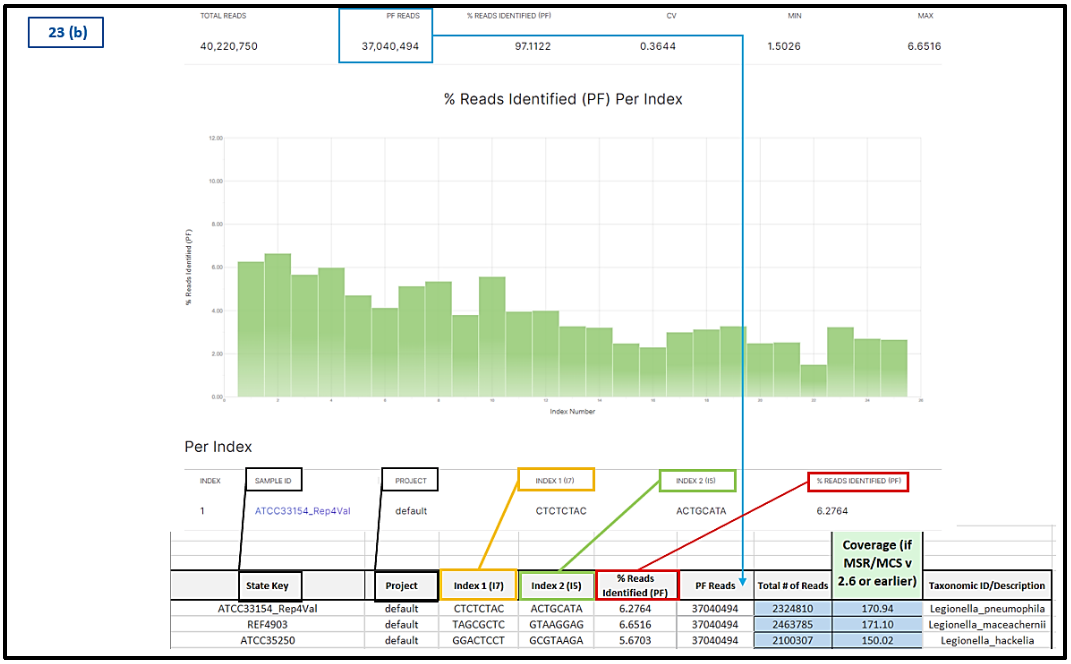


D. Fill out the workbook metrics:

1. **Metrics tab**: Copy run metrics from BaseSpace into the relevant cells of the Metrics tab of the WGS worksheet.
    i. Cluster Density
    ii. %Q30
    iii. Cluster Passing Filter (%PF)
    iv. Estimated Yield (Gb)



2. **Indexing QC tab**:
    i. In the "per index" section, copy all the data under Index through %Reads Identified (PF).
    ii. Paste into the first row under Index Number in the Metrics tab of the WGS worksheet.
    iii. Copy the PF reads value from BaseSpace into the PF reads column of the worksheet.



The worksheet will automatically calculate total reads and coverage (i.e. sequencing depth) if you have entered the estimated genome size and chemistry you used correctly.

Note: **The following fields must be filled out correctly for the coverage to be calculated accurately: LP tab (Sample ID, Genome Size in Mb, and Sequencing Kit Type/Chemistry), and the Metrics tab (Sample ID should match the LP tab, the % reads identified, and PF reads).**

11 POLLUTION PREVENTION/WASTE MANAGEMENT

A. No infectious waste is generated in performing this protocol.

B. All hazardous waste is disposed of according to current PHL standards.

- **Important:** All liquid that accumulates in the MiSeq waste bottle should be disposed of as formamide waste, including liquid from wash cycles. All formamide waste bottles must be clearly labeled with contents, NFPA labels, date of first use, and date of final use.
- Waste must be stored in a designated, clearly labeled waste disposal site.
- A satellite container in R-22 is used to collect routine waste from MiSeq and transferred to long term storage, and disposal. Once the satellite container is 75% full, it is transferred to a larger container in the hazardous storage area (B-106) and stored there until picked up by the hazardous waste management company.
- Liquid from the waste bottles and from position 8 in the MiSeq cartridge must be discarded in the formamide waste satellite container while working in the fume hood in R-22.

C. Place all other waste in a bucket lined with a biohazard bag. Once the bag is 75% full, fold the bag down, tape, and place in a coffin lined with a biohazard bag.

D. Once the coffin is $^{2}/_{3}$ full, add 250mL of water and fold the biohazard bag twice, and tape.

E. Tape the lid with a biohazardous-waste label. Place the coffin on an autoclave cart in the OPS dish room.

F. Waste that should be placed in a coffin includes pipette tips, micro centrifuge tubes, gloves, etc.

12 INTERPRETATIONS

A. Library Preparation

1. The fragment peak length should be approximately between 250 – 350bp with insert sizes ranging from approximately 100 – 200bp.

- Longer fragments can lead to low cluster density possibly resulting in inadequate coverage.
- Shorter fragments can lead to high cluster density and can also result in inadequate coverage (fewer clusters can pass the filter) and/or poor-quality data.
- Shorter than 200 bp fragment indicates failed library preparation. Consult with an AMD Lead or Supervisor to reattempt partial or complete library preparation.

B. Loading an NGS Run on the MiSeq

Once the run is complete the quality metrics can be verified using BaseSpace prior to giving the report for interpretation.

1. BaseSpace
   - **Review run metrics**: Upon run completion, confirm that the sequencing run meets the basic quality metrics (Q30, Cluster Density, and Cluster Passing Filter).

Kit Chemistry	Q30%	Cluster Density ($^{k}/_{mm^2}$)	Cluster Passing Filter (%)
v2, 300c	≥ 80%	800 – 1400	> 80%

Note: **If the %Q30, the Cluster Density, and Cluster Passing Filter metrics do not meet the threshold above, the run may or may not need to be repeated. Further analysis of the sequence data requires consultation with a subject matter expert.**

2. **Error log instructions:** Non-conforming events should be discussed with Molecular Lead and/or Supervisor and documented. Refer to MediaLab Document #: 69790.945.
3. **Corrective Action:** Corrective action for known events resulting in error should be reviewed by Molecular Lead or Supervisor. All troubleshooting steps must be approved by Molecular Lead/Supervisor and documented for QI purposes.

C. MiSeq Data QC and Reporting

If the run does not pass the above QC, it may not be sufficient to proceed with downstream bioinformatic analysis QC. Consult with a lead microbiologist. For detailed bioinformatic QC criteria, please refer to **MediaLab 69790.5317 SOP: Viral Assembly and Taxonomic Classification with VAPER.**

13 CALCULATIONS

The workbook and checklist will calculate almost all calculations needed for a successful library prep. Calculations in the workbook are locked for editing. Consult with Molecular Lead/Supervisor for editing.

A. To calculate Molarity (M):

$$\text{Molarity (nM)} = \frac{\text{Pool Concentration} \times 1000000}{660^{g}/_{mol} \times \text{insert size}}$$

B. To calculate a 6nM pooled library (6nM):

$$\text{Volume of Library Pool to Dilute } (\mu L) = \frac{\text{Final concentration (6nM)} \times 50\mu L}{\text{Molarity (nM)}}$$

$$\text{Volume of RSB to Dilute } (\mu L) = 50\mu L - \text{Volume of Library Pool to Dilute } (\mu L)$$

14 REFERENCE RANGE

The protocol is optimized for the following input sample types:

A. Clinical and environmental isolates containing viruses with dsDNA/ssRNA/ssDNA genomes stored in VTM/UTM/PBS or other transport media.

B. Total nucleic acid extracts from clinical and environmental isolates containing viruses with dsDNA/ssRNA/ssDNA genomes.

The protocol is optimized for final sequencing libraries with fragment length: ~250 – 350bp. Lower input amounts and lesser quality can reduce library yield. Due to the highly complex nature of real-world samples, using equal volume of extracted nucleic acid per sample is preferred.

15 REPORTING RESULTS

A. Record all data and lot numbers in the Workbook and the Reagent tracker:

1. A report is made to document each run and sample QC information.
2. Update the WGS Tracker and note any pertinent information (QC failed, repeats, etc.).
3. The version number of the Bioinformatics pipeline used to analyze the run will be recorded on the QC cover page and will be updated as needed.
4. Highlight repeats in red, and high priorities in pink on the Metrics page. Note any pertinent information in the respected comment section within the cover page.

B. Assemble the report:

1. Print the sheets described in the QC cover page of the workbook
2. Give report to the Lead Microbiologist or Supervisor for technical review.
3. **Once sequence data review is complete and passed**, the run packet will be archived at AMD both in printed and/or digital formats. The Sequencing data will be transferred to the bioinformatics team for downstream analysis.
4. **If the run fails**, the report is still made to document the failure. This report is given to the Lead Microbiologist for review.

16 REFERENCES

A. Illumina Viral Surveillance Panel v2 kit Reference Guide (Media Lab Document #: 69790.5098).

B. Promega Maxwell® CSC 48 Instrument IVD Mode Operating Manual TM623 (Media Lab Document #: 69790.5006).

C. Molecular Devices SpectraMax M2 Microplate Reader (Media Lab Document #: 69790.1624).

D. Best Practices Review: Bead Handling Illumina (Media Lab Document #: 69790.964).

E. Agilent TapeStation D5000 High Sensitivity Quick Guide (Media Lab Document #: 69790.2293).

F. Agilent TapeStation 4200 Manual (Media Lab Document #: 69790.1614).

G. Qubit dsDNA HS Assay (Media Lab Document #: 69790.3132).

H. Illumina MiSeq System Guide (Media Lab Document #: 69790.3163).

17 APPENDICES

17.1 Appendix A. Abbreviations, Definitions, and Symbols Used

a. **WGS:** Whole Genome Sequencing
b. **NGS:** Next-Generation Sequencing
c. **VSP:** Viral Surveillance Panel
d. **TNA:** Total Nucleic Acid
e. **RNA:** Ribonucleic Acid
f. **cDNA:** Complementary DNA
g. **EBLTL:** Enrichment Bead-Linked Transposomes
h. **UDI:** Unique Dual Indexes
i. **IPB:** Illumina Purification Beads
j. **ST2:** Stop Tagmentation Buffer 2
k. **TWB:** Tagmentation Wash Buffer
l. **RSB:** Resuspension Buffer
m. **EHB2:** Enrich Hyb Buffer 2
n. **ET2:** Elute Target Buffer 2
o. **SMB3:** Streptavidin Magnetic Beads 3
p. **TB1:** Tagmentation Buffer 1
q. **EPM:** Enhanced PCR Mix
r. **EE1:** Enrichment Elution Buffer 1
s. **EEW:** Enhanced Enrichment Wash
t. **HP3:** 2N NaOH
u. **NHB2:** Hyb Buffer 2 + IDT NXT Blockers
v. **PPC:** PCR Primer Cocktail
w. **EPH3:** Elute, Prime, Fragment High Concentration Mix
x. **FSA:** First Strand Synthesis Act D Mix
y. **RVT:** Reverse Transcriptase
z. **SMM:** Second Strand Marking Master Mix
aa. **PR2:** Incorporation Buffer
bb. **TAT:** Turn Around Time
cc. **DNA:** Deoxyribonucleic Acid
dd. **dsDNA:** double-stranded DNA
ee. **HS:** High Sensitivity
ff. **PCR:** Polymerase Chain Reaction
gg. **PHL:** Public Health Laboratory
hh. **PPE:** Personal Protective Equipment
ii. **QC:** Quality Control
jj. **RNase:** Ribonuclease
kk. **SDS:** Safety Data Sheets
ll. **NTC:** Non-Template Control
mm. **WAPHL:** Washington State Department of Health

17.2 Appendix B. Supporting Documents

For Type indicate if it is a SOP, Form, Validation, Job Aid, Product Insert, Manual, etc.

MediaLab Document #	Title	Type
69790.17	Biosafety Manual	Manual
69790.38	Safety and Health Manual	Manual
69790.945	How to Record a New Nonconforming Event (NCE) in MediaLab	SOP
69790.956	WAPHL DNA Quantification and QC	SOP
69790.964	BeadHandling BestPractices	Manual
69790.966	Qubit HS kit Concentration Measurement	Job Aid
69790.1229	Laboratory Contamination Prevention and Monitoring: Nucleic Acid Testing	SOP
69790.1546	PM Form – MiSeq Maintenance	Form
69790.1614	Agilent TapeStation 4200 Manual	Manual
69790.1624	SpectraMax M2/M2e User Guide	Manual
69790.2293	Agilent Tapestation D5000 High Sensitivity QuickGuide	Manual
69790.2678	AMD Risk Assessment	Risk Assessment
69790.3106	MiSeq Wash and Instrument Maintenance	Job Aid
69790.3108	PhiX Preparation	Job Aid
69790.3109	Flow Rate Failure	Job Aid
69790.3110	Re-queueing Run Analysis	Job Aid
69790.3132	Qubit dsDNA HS Assay	Manual
69790.3163	Illumina MiSeq System Guide	Manual
69790.3668	Measuring Insert Size with TapeStation	Job Aid
69790.4374	Illumina Sequencing Technology	Manual
69790.4364	Creating a Sample Sheet using Local Run Management	Job Aid
69790.4338	Illumina Purification Beads Performance Comparison	Validation Report
69790.4489	New MiSeq Performance Comparison	Validation Report
69790.5006	Promega Maxwell® CSC 48 Instrument IVD Mode Operating Manual TM623	Manual
69790.5098	Illumina Viral Surveillance Panel v2 kit Reference Guide	Manual
69790.5386	Job Aid - Promega Maxwell Viral Total Nucleic Acid Extraction	Job Aid
69790.5258	Illumina Viral Surveillance Library Preparation Workbook	Workbook
69790.5317	SOP: Viral Assembly and Taxonomic Classification with VAPER	SOP
69790.5094	Viral Genomic Sequencing using Illumina Viral Surveillance Kit	SOP
69790.5283	Validation Plan - Sequencing and Bioinformatic Analysis of Viral Genomic Sequencing using Illumina Viral Surveillance Kit	Validation Plan
69790.5387	Illumina Viral Surveillance Sequencing Validation Report	Validation Report

17.3 Appendix C. Viral Targets for the Viral Surveillance Panel v2 Kit.

Reporting Names	Reporting Names	Reporting Names	Reporting Names
Adeno-associated virus 2 (AAV2)	Human coronavirus 229E (HCoV_229E)	Influenza A virus (H7N3)	Ravn virus (RAVV)
Aichi virus 1 (AiV-A1)	Human coronavirus HKU1 (HCoV_HKU1)	Influenza A virus (H7N4)	Reston virus (RESTV)
Aigai virus (AIGV)	Human coronavirus NL63 (HCoV_NL63)	Influenza A virus (H7N7)	Rhinovirus A (RV-A)
Andes virus (ANDV)	Human coronavirus OC43 (HCoV_OC43)	Influenza A virus (H7N9)	Rhinovirus B (RV-B)
Anjozorobe virus (ANJV)	Human cytomegalovirus (HCMV)	Influenza A virus (H9N2)	Rhinovirus C (RV-C)
Araucaria virus	Human immunodeficiency virus 1 (HIV-1)	Influenza B virus	Rift Valley fever virus (RVFV)
Australian bat lyssavirus (ABLV)	Human immunodeficiency virus 2 (HIV-2)	Influenza B virus (B/Victoria/2/87-like)	Ross River virus (RRV)
Bayou virus (BAYV)	Human metapneumovirus (HMPV)	Influenza B virus (B/Yamagata/16/88-like)	Rotavirus A (RVA)
BK polyomavirus (BKPyV)	Human papillomavirus 11 (HPV11)	Influenza C virus	Rotavirus B (RVB)
Black Creek Canal virus (BCCV)	Human papillomavirus 16 (HPV16; high-risk)	Isla Vista virus	Rotavirus C (RVC)
Bombali virus (BOMV)	Human papillomavirus 18 (HPV18; high-risk)	Itapua virus	Rotavirus H (RVH)
Bourbon virus (BRBV)	Human papillomavirus 26 (HPV26)	Jamestown Canyon virus (JCV)	Rubella virus (RuV)
Bundibugyo virus (BDBV)	Human papillomavirus 31 (HPV31; high-risk)	Japanese encephalitis virus (JEV)	Sabia virus (SBAV)
Cache Valley virus (CVV)	Human papillomavirus 33 (HPV33; high-risk)	JC polyomavirus (JCPyV)	Salivirus A (SaV-A)
California encephalitis virus (CEV)	Human papillomavirus 35 (HPV35; high-risk)	Junin virus (JUNV)	Sandfly fever Sicilian virus (SFCV)
Cedar virus (CedV)	Human papillomavirus 39 (HPV39; high-risk)	Juquitiba virus	Sangassou virus (SANGV)
Chapare virus (CHAPV)	Human papillomavirus 40 (HPV40)	KI polyomavirus (KIPyV)	Sapovirus
Chikungunya virus (CHIKV)	Human papillomavirus 42 (HPV42)	Kyasanur Forest disease virus (KFDV)	Semliki Forest virus (SFV)
Choclo virus (CHOV)	Human papillomavirus 43 (HPV43)	La Crosse virus (LACV)	Seoul virus (SEOV)
Colorado tick fever virus (CTFV)	Human papillomavirus 44 (HPV44)	Lagos bat virus (LBV)	Severe acute respiratory syndrome coronavirus (SARS-CoV)
Coxsackievirus A (species Enterovirus A)	Human papillomavirus 45 (HPV45; high-risk)	Laguna Negra virus (LANV)	Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)
Coxsackievirus A (species Enterovirus B)	Human papillomavirus 51 (HPV51; high-risk)	Langya virus	Severe fever with thrombocytopenia syndrome virus (SFTSV)
Coxsackievirus A (species Enterovirus C)	Human papillomavirus 52 (HPV52; high-risk)	Lassa virus (LASV)	Simian virus 40 (SV40)
Coxsackievirus B (species Enterovirus B)	Human papillomavirus 53 (HPV53)	LI polyomavirus (LIPyV)	Sin nombre virus (SNV)
Crimean-Congo hemorrhagic fever virus (CCHFV)	Human papillomavirus 54 (HPV54)	Lloviu virus (LLOV)	Sindbis virus (SINV)
Dengue virus (DENV)	Human papillomavirus 56 (HPV56; high-risk)	Lujo virus (LUJV)	Snowshoe hare virus (SSHV)
Dengue virus type 1 (DENV-1)	Human papillomavirus 58 (HPV58; high-risk)	Luxi virus (LUXV)	Sosuga virus (SoRV)
Dengue virus type 2 (DENV-2)	Human papillomavirus 59 (HPV59; high-risk)	Lymphocytic choriomeningitis virus (LCMV)	St. Louis encephalitis virus (SLEV)
Dengue virus type 3 (DENV-3)	Human papillomavirus 6 (HPV6)	Machupo virus (MACV)	STL polyomavirus (STLPyV)
Dengue virus type 4 (DENV-4)	Human papillomavirus 61 (HPV61)	Mamastrovirus 1 (MAstV1)	Sudan virus (SUDV)
Dobrava virus (DOBV)	Human papillomavirus 66 (HPV66; high-risk)	Mamastrovirus 6 (MAstV6)	Tacheng tick virus 2 (TcTV-2)
Duvenhage virus (DUVV)	Human papillomavirus 68 (HPV68; high-risk)	Mamastrovirus 8 (MAstV8)	Tahyna virus (TAHV)
Eastern equine encephalitis virus (EEEV)	Human papillomavirus 69 (HPV69)	Mamastrovirus 9 (MAstV9)	Tai Forest virus (TAFV)
Ebola virus (EBOV)	Human papillomavirus 70 (HPV70)	Maporal virus (MAPV)	Tick-borne encephalitis virus (TBEV)
Echovirus (species Enterovirus B)	Human papillomavirus 73 (HPV73)	Marburg virus (MARV)	Torque teno virus (TTV)
Enterovirus A	Human papillomavirus 82 (HPV82)	Mayaro virus (MAYV)	Toscana virus (TOSV)
Enterovirus A (not Coxsackievirus)	Human parainfluenza virus 1 (HPIV-1)	Measles virus (MV)	Trichodysplasia spinulosa-associated polyomavirus (TSPyV)
Enterovirus B	Human parainfluenza virus 2 (HPIV-2)	Menangle virus (MenV)	Tula virus (TULV)
Enterovirus B (not Coxsackievirus or Echovirus)	Human parainfluenza virus 3 (HPIV-3)	Merkel cell polyomavirus (MCPyV)	Usutu virus (USUV)
Enterovirus C	Human parainfluenza virus 4 (HPIV-4)	Middle East respiratory syndrome-related coronavirus (MERS-CoV)	Varicella-zoster virus (VZV)
Enterovirus C (not Coxsackievirus or Poliovirus)	Human parechovirus (HPeV)	Mojiang virus (MojV)	Variola virus (VARV)
Enterovirus D	Human parvovirus B19 (B19V)	Mokola virus (MOKV)	Venezuelan equine encephalitis virus (VEEV)
Epstein-Barr virus (EBV)	Human polyomavirus 6 (HPyV6)	Monkeypox virus (MPV)	West Nile virus (WNV)
European bat lyssavirus (EBLV)	Human polyomavirus 7 (HPyV7)	Monongahela hantavirus	Western equine encephalitis virus (WEEV)
Ghana virus (GhV)	Human polyomavirus 9 (HPyV9)	Muleshoe virus	WU polyomavirus (WUPyV)
Guanarito virus (GTOV)	Human respiratory syncytial virus A (HRSV-A)	Mumps virus (MuV)	Yellow fever virus (YFV)
Hantaan virus (HTNV)	Human respiratory syncytial virus B (HRSV-B)	Murray Valley encephalitis virus (MVEV)	Zika virus (ZIKV)
Heartland virus (HRTV)	Influenza A virus	MW polyomavirus (MWPyV)	
Hendra virus (HeV)	Influenza A virus (H10N3)	New Jersey polyomavirus (NJPyV)	
Henipavirus unclassified	Influenza A virus (H10N7)	Nipah virus (NiV)	
Hepatitis A virus (HAV)	Influenza A virus (H10N8)	Norovirus	
Hepatitis B virus (HBV)	Influenza A virus (H1N1)	Norovirus GI	
Hepatitis C virus (HCV)	Influenza A virus (H1N1; swine- or avian-like)	Norovirus GII	
Hepatitis D virus (HDV)	Influenza A virus (H1N1; pdm09-like)	Norovirus GIV	
Hepatitis E virus (HEV)	Influenza A virus (H1N2)	Norovirus GIX	
Herpes simplex virus 1 (HSV1)	Influenza A virus (H2N2)	Norovirus GVIII	
Herpes simplex virus 2 (HSV2)	Influenza A virus (H3N2)	Omsk hemorrhagic fever virus (OHFV)	
Human adenovirus A	Influenza A virus (H3N2; avian-like)	Onyong-nyong virus (ONNV)	
Human adenovirus B	Influenza A virus (H3N2; swine-like)	Oropouche virus (OROV)	
Human adenovirus C	Influenza A virus (H3N2; human-like)	Paranoa virus	
Human adenovirus D	Influenza A virus (H5N1)	Poliovirus (species Enterovirus C)	
Human adenovirus E	Influenza A virus (H5N6)	Powassan virus (POWV)	
Human adenovirus F	Influenza A virus (H5N8)	Punta Toro virus (PTV)	
Human adenovirus G	Influenza A virus (H6N1)	Puumala virus (PUUV)	
Human bocavirus (HBoV)	Influenza A virus (H7N2)	Rabies virus (RABV)	